

# CFD ANALYSIS OF AIRFLOW AND PRESSURE LOSS IN VENTILATION DUCT GEOMETRIES

---

*Comparative Study of Sharp-Edge and Curved-Bend Duct Designs*

Simulation Tool: Autodesk CFD 2026

Turbulence Model: k- $\epsilon$  (Standard)

Inlet Velocity: 5 m/s | Fluid: Air (Incompressible)

**January 2026 – February 2026**

## 1. Abstract

This report presents a Computational Fluid Dynamics (CFD) analysis of airflow characteristics and pressure loss in two ventilation duct geometries: a sharp 90° bend and a curved (radius) bend. Both geometries were designed in Fusion 360 and simulated using Autodesk CFD 2026 under identical boundary conditions. The study evaluates velocity distribution, turbulence behavior, streamline patterns, and pressure loss across the bend region. Results demonstrate that the curved bend achieves a significantly lower pressure loss of 21.44 Pa compared to 34.91 Pa for the sharp bend, representing an improvement of approximately 38.6%. Flow separation and recirculation, clearly observed in the sharp bend, are substantially reduced in the curved design, confirming that curved bends offer superior aerodynamic performance for ventilation applications.

## 2. Introduction

In HVAC (Heating, Ventilation, and Air Conditioning) systems, duct geometry plays a critical role in determining energy efficiency and airflow quality. Bends or elbows in duct systems are common sources of pressure loss, turbulence, and flow non-uniformity. Understanding how geometry influences these parameters is essential for designing efficient ventilation systems.

This study uses CFD simulation to compare two duct bend configurations: a sharp 90° corner and a smooth curved bend. The objectives are to visualize velocity and pressure distributions, identify regions of flow separation, and quantify pressure losses to assess which geometry provides better airflow efficiency.

## 3. Methodology

### 3.1 Geometry Design

Two duct geometries were modeled in Autodesk Fusion 360 and exported as STEP (.step) files for import into Autodesk CFD 2026. Both ducts share identical cross-sectional dimensions and inlet/outlet lengths, differing only in the bend region:

- Sharp Bend Duct: A 90° elbow with a sharp inner corner (zero bend radius), representing a worst-case geometry for flow disturbance.
- Curved Bend Duct: A 90° elbow with a smooth toroidal bend radius of 125 mm, which promotes gradual flow redirection.

### 3.2 CFD Setup and Boundary Conditions

The following setup was applied identically to both simulations:

| Parameter    | Value             |
|--------------|-------------------|
| CFD Software | Autodesk CFD 2026 |

| Parameter                 | Value                                   |
|---------------------------|-----------------------------------------|
| Fluid Material            | Air (Incompressible)                    |
| Air Density               | 1.225 kg/m <sup>3</sup>                 |
| Dynamic Viscosity         | 1.81 × 10 <sup>-5</sup> Pa·s            |
| Inlet Boundary Condition  | Velocity Inlet – 5 m/s (normal to face) |
| Outlet Boundary Condition | Pressure Outlet – 0 Pa (gauge)          |
| Wall Condition            | No-slip (default)                       |
| Turbulence Model          | k-ε (Standard)                          |
| Flow Type                 | Turbulent, Steady State                 |
| Solution Mode             | Incompressible                          |
| Heat Transfer             | Off                                     |
| Iterations                | 300–800 (until convergence)             |

### 3.3 Convergence Criteria

Convergence was assessed by monitoring residuals for velocity components ( $V_x$ ,  $V_y$ ,  $V_z$ ), pressure, turbulent kinetic energy (TKE), and turbulent energy dissipation (TED). The solver was deemed converged when all residual lines flattened — confirmed by the Autodesk CFD “Flat Lines Detected” notification. The sharp bend required approximately 480 iterations, while the curved bend converged earlier due to smoother flow behavior.

## 4. Results

### 4.1 Sharp Bend — Velocity Distribution ( $V_y$ Component)

The  $V_y$  velocity contour plot (Figure 1) reveals significant flow disturbance at the inner corner of the sharp bend. High-velocity zones (red/orange, up to 738.5 cm/s) concentrate at the outer wall of the bend, while a prominent low-velocity recirculation zone (blue) forms immediately after the sharp inner corner. This flow separation persists well into the outlet section, causing non-uniform velocity distribution downstream.

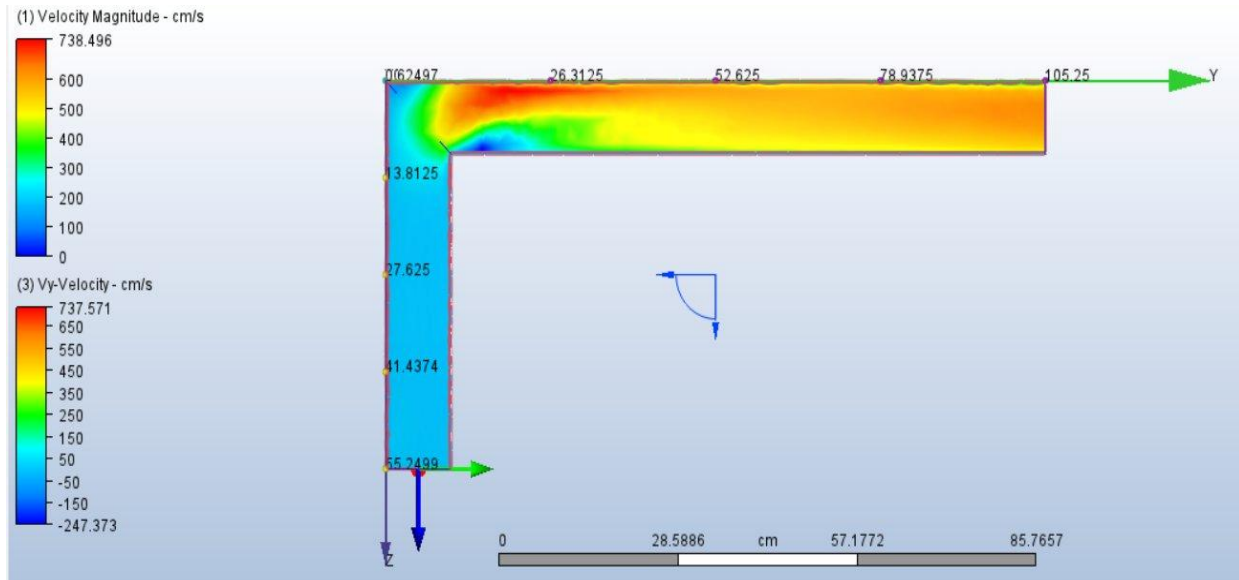


Figure 1: Sharp Bend —  $V_y$  Velocity Contour (cm/s). Max velocity: 738.5 cm/s. Clear flow separation visible at inner corner.

#### 4.2 Curved Bend — Velocity Distribution ( $V_y$ Component)

The curved bend (Figure 2) shows a markedly smoother velocity distribution. The maximum velocity (645.2 cm/s) is lower than the sharp bend, and the high-velocity zone transitions gradually around the bend without abrupt separation. The flow remains largely attached to the inner wall throughout the curve, resulting in a more uniform outlet velocity profile.

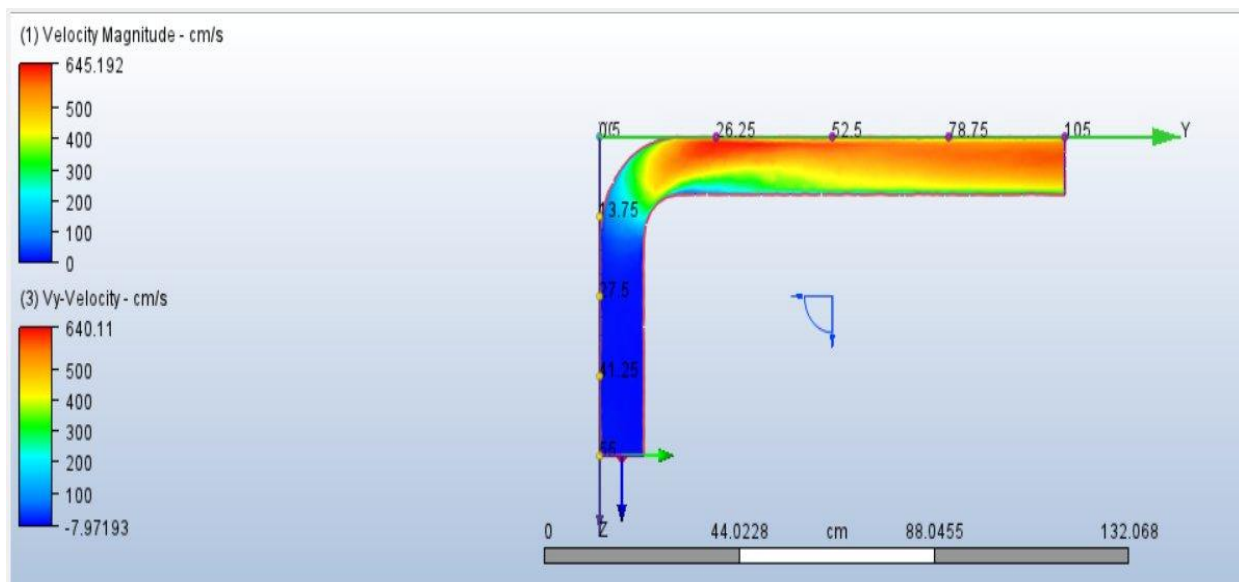


Figure 2: Curved Bend —  $V_y$  Velocity Contour (cm/s). Max velocity: 645.2 cm/s. Smooth flow transition with minimal separation.

#### 4.3 Sharp Bend — Velocity Distribution ( $V_z$ Component)

The  $V_z$  velocity component (Figure 3) further highlights the three-dimensional nature of the flow disturbance in the sharp bend. Negative  $V_z$  values (blue, down to -644.4 cm/s) indicate reversed

flow components in the Z-direction near the separation zone, confirming the presence of a strong recirculation region.

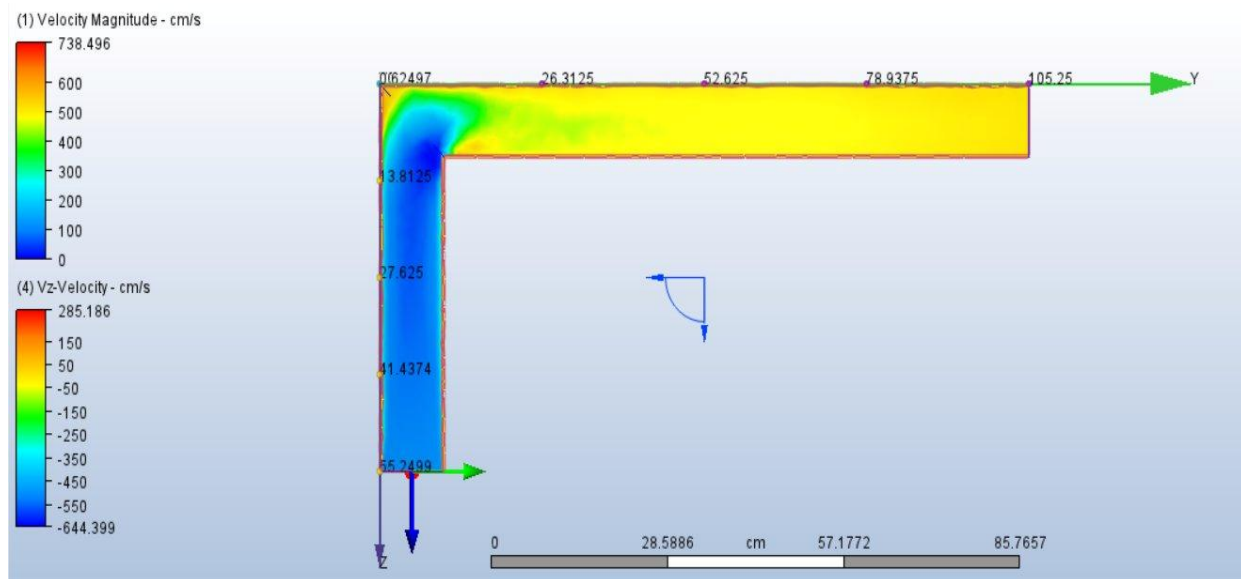


Figure 3: Sharp Bend — Vz Velocity Component (cm/s). Negative values indicate flow reversal and recirculation at the bend.

#### 4.4 Curved Bend — Velocity Distribution (Vz Component)

In contrast, the curved bend Vz distribution (Figure 4) shows significantly reduced negative velocity regions (minimum -612.6 cm/s concentrated only at the bend transition), confirming that the curved geometry substantially suppresses reversed flow and three-dimensional turbulence effects.

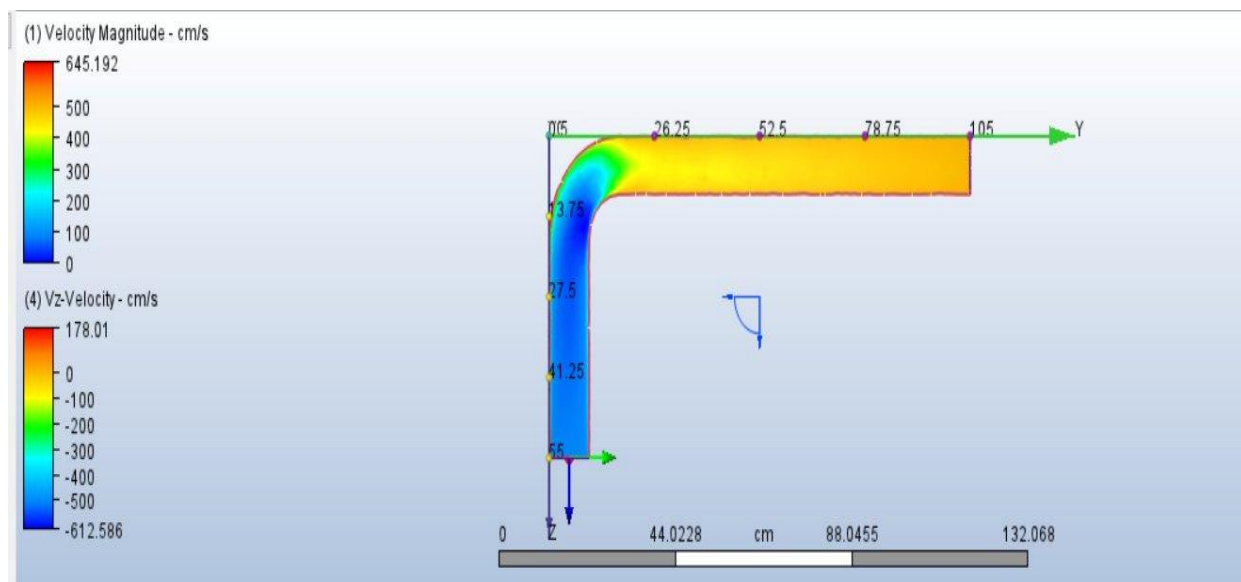


Figure 4: Curved Bend — Vz Velocity Component (cm/s). Reduced flow reversal compared to sharp bend.

#### 4.5 Sharp Bend — Static Pressure Distribution

The static pressure map of the sharp bend (Figure 5) shows a steep pressure gradient across the bend. High pressure (orange/red, up to 409.3 dyne/cm<sup>2</sup>) accumulates at the inlet and outer bend wall as flow is forced to change direction abruptly. The pressure drops sharply to near-zero and negative values (-34.93 dyne/cm<sup>2</sup>) at the inner corner — indicative of local flow acceleration and separation. The measured total pressure loss between inlet and outlet was 34.91 Pa.

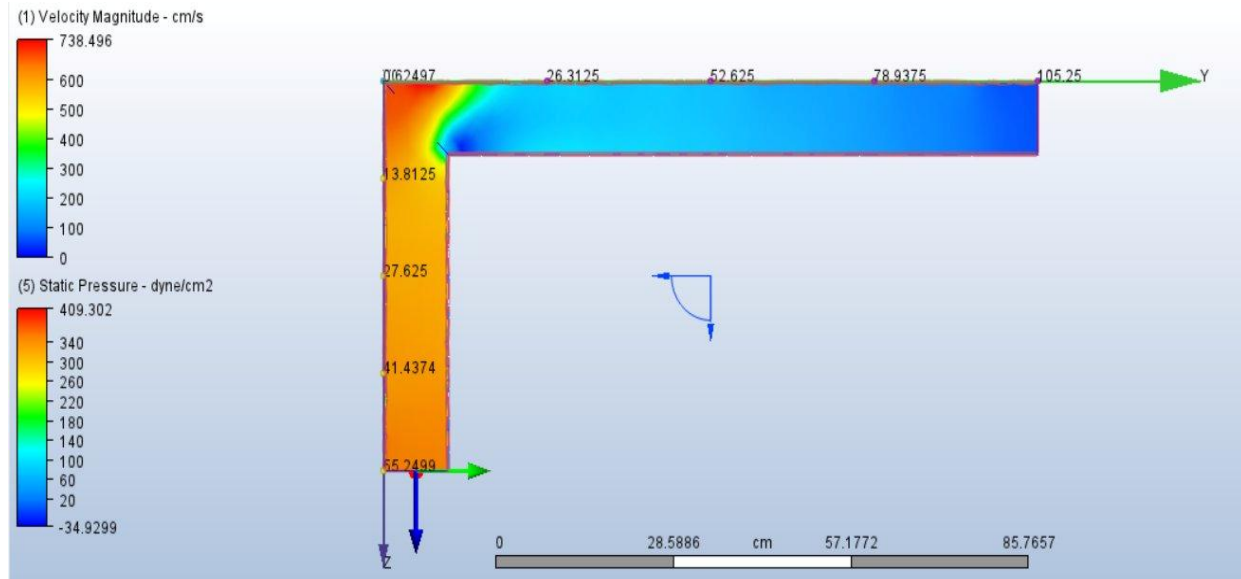
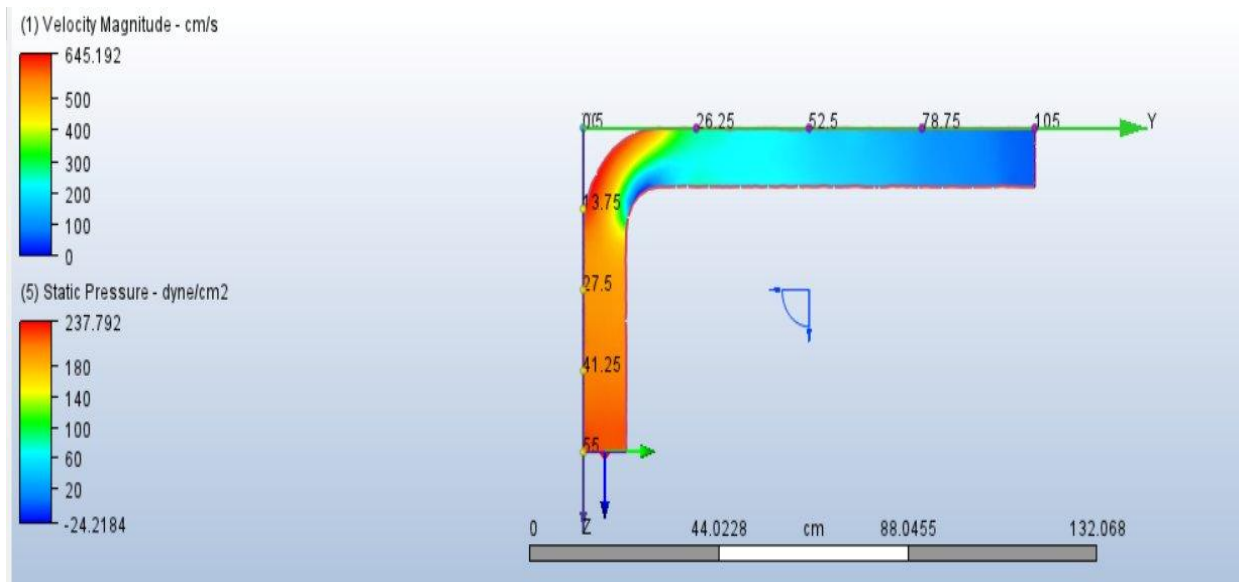


Figure 5: Sharp Bend — Static Pressure Distribution (dyne/cm<sup>2</sup>). High inlet pressure, steep drop at bend. Total pressure loss: 34.91 Pa.

#### 4.6 Curved Bend — Static Pressure Distribution

The curved bend pressure distribution (Figure 6) demonstrates a more gradual and uniform pressure decrease from inlet to outlet. The maximum inlet pressure (237.8 dyne/cm<sup>2</sup>) is significantly lower than the sharp bend, and the pressure gradient is smoother throughout the bend region. Negative pressures are minimal (-24.2 dyne/cm<sup>2</sup>), indicating reduced flow separation. The total measured pressure loss was 21.44 Pa.





The curved bend streamlines (Figure 8) reveal a dramatically different flow pattern. Streamlines follow the bend curvature smoothly without detachment, maintaining relatively uniform spacing throughout. There is no visible recirculation zone, and velocity stratification is minimal. The flow recovers to a nearly uniform profile well within the outlet section.

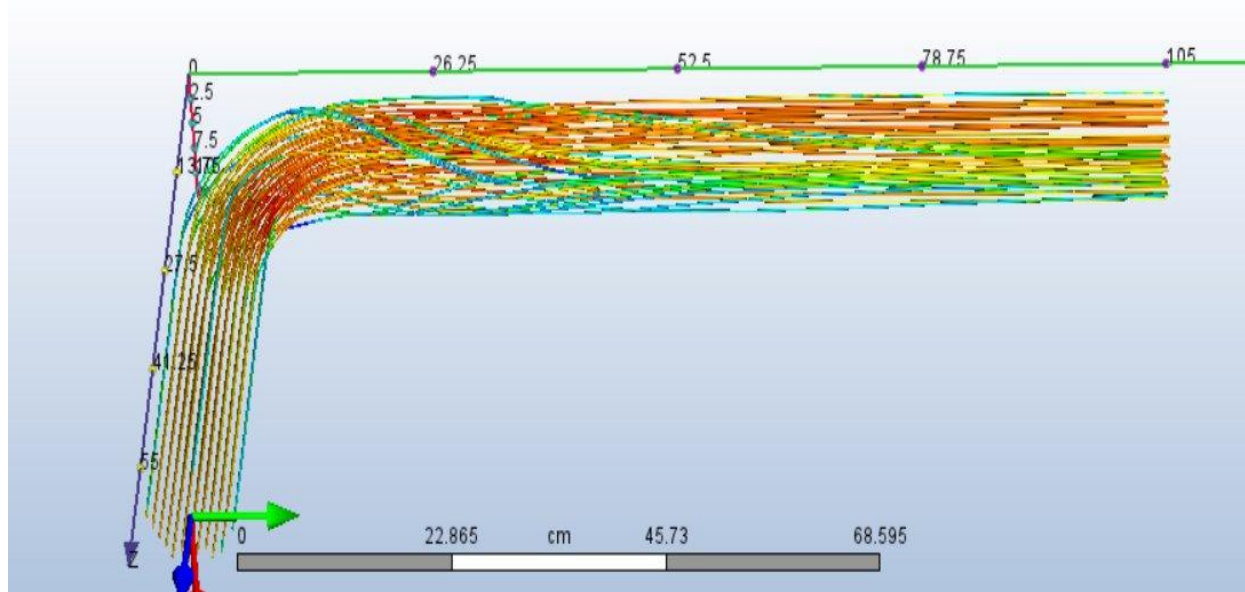


Figure 8: Curved Bend — Streamline Traces. Smooth, attached flow with no significant separation or recirculation.

## 5. Comparative Analysis

| Parameter              | Sharp Bend                  | Curved Bend                 | Improvement          |
|------------------------|-----------------------------|-----------------------------|----------------------|
| Inlet Velocity         | 5 m/s                       | 5 m/s                       | —                    |
| Max Velocity ( $V_y$ ) | 738.5 cm/s                  | 645.2 cm/s                  | 12.6% lower peak     |
| Max Static Pressure    | 409.3 dyne/cm <sup>2</sup>  | 237.8 dyne/cm <sup>2</sup>  | 41.9% reduction      |
| Min Static Pressure    | -34.93 dyne/cm <sup>2</sup> | -24.22 dyne/cm <sup>2</sup> | 30.7% less negative  |
| Total Pressure Loss    | 34.91 Pa                    | 21.44 Pa                    | 38.6% reduction      |
| Flow Separation        | Severe (inner corner)       | Minimal                     | ✓ Eliminated         |
| Recirculation Zone     | Present                     | Absent                      | ✓ Eliminated         |
| Outlet Uniformity      | Non-uniform                 | Largely uniform             | ✓ Improved           |
| Iterations to Converge | ~480                        | Earlier (flat lines)        | ✓ Faster convergence |



## 6. Discussion

---

The results clearly demonstrate the aerodynamic superiority of the curved bend over the sharp bend design across all measured parameters. The 38.6% reduction in pressure loss from 34.91 Pa to 21.44 Pa is the most significant finding, directly translating to lower energy consumption in real ventilation systems.

The sharp bend forces air to change direction abruptly at 90°, creating an adverse pressure gradient at the inner corner. The flow cannot follow this sharp geometry change (a phenomenon known as flow separation), forming a dead zone of recirculating air that occupies a significant portion of the duct cross-section. This recirculation zone effectively reduces the usable flow area and generates additional turbulent losses. The downstream velocity remains non-uniform for a considerable length, meaning more duct length is required before flow stabilizes.

The curved bend, by providing a gradual direction change, allows the flow to remain attached to the duct walls throughout the bend. The momentum of the fluid is redirected more efficiently, with significantly less turbulence generation. The lower peak velocity (645.2 cm/s vs. 738.5 cm/s) indicates that the flow does not need to accelerate as dramatically to navigate the bend, which is consistent with lower dynamic pressure losses.

These findings align with established fluid dynamics principles and support the use of curved bends (with radius-to-diameter ratios of 1.5 or greater) in HVAC duct design, particularly where energy efficiency and noise reduction are priorities.

## 7. Conclusion

---

- Both duct geometries were successfully simulated using Autodesk CFD 2026 with identical boundary conditions (5 m/s inlet velocity, k-ε turbulence model, incompressible steady-state flow).
- The sharp bend produced a total pressure loss of 34.91 Pa, accompanied by severe flow separation, a recirculation dead zone at the inner corner, and non-uniform outlet velocity distribution.
- The curved bend produced a total pressure loss of 21.44 Pa — a 38.6% reduction — with smooth, attached flow, no recirculation, and a significantly more uniform outlet velocity profile.
- Streamline visualizations confirmed that the curved bend eliminates the separated flow region that characterizes the sharp bend design.
- For ventilation duct systems, curved bends are strongly recommended over sharp bends to minimize pressure losses, reduce turbulence, and improve overall system energy efficiency.

## 8. References & Tools

---

- Autodesk CFD 2026 — Simulation and post-processing platform
- Autodesk Fusion 360 — Geometry design and STEP file export
- ASHRAE Handbook — Fundamentals: Duct Design and Pressure Loss Coefficients

- Anderson, J.D. (2011). Computational Fluid Dynamics: The Basics with Applications. McGraw-Hill.
- Versteeg, H.K. & Malalasekera, W. (2007). An Introduction to Computational Fluid Dynamics: The Finite Volume Method. Pearson.